P8109 Statistical Inference – Homework 2

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Problem 1

Let $X_1, \ldots, X_n$ be i.i.d. Poisson(λ) random variables. Find the MLE of λ.

Solution. The likelihood function is

$$L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{X_i}}{X_i!} = \frac{e^{-n\lambda}\lambda^{\sum_{i=1}^{n} X_i}}{\prod_{i=1}^{n} X_i!}.$$

The log-likelihood is

$$\ell(\lambda) = -n\lambda + \left(\sum_{i=1}^{n} X_i\right)\log\lambda - \sum_{i=1}^{n}\log(X_i!).$$

Taking the derivative and setting it to zero:

$$\frac{d\ell}{d\lambda} = -n + \frac{\sum_{i=1}^{n} X_i}{\lambda} = 0.$$

Solving gives $\boxed{\hat{\lambda}_{\mathrm{MLE}} = \bar{X}}$.

The second derivative $d^2\ell/d\lambda^2 = -\sum X_i/\lambda^2 < 0$ confirms this is a maximum.

Problem 2

Consider a sequence of i.i.d. random variables from a $N(\mu, \sigma^2)$ distribution. Let $m = \frac{1}{n}\sum_{i=1}^{n} X_i^2$.

(a) Given that $\sigma^2 = \mu$, show that the MLE of μ is $\hat{\mu} = \frac{-1+\sqrt{1+4m}}{2}$.

Proof. When $\sigma^2 = \mu$, the log-likelihood is

$$\ell(\mu) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\mu - \frac{1}{2\mu}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Expanding $(X_i - \mu)^2 = X_i^2 - 2\mu X_i + \mu^2$:

$$\ell(\mu) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\mu - \frac{nm}{2\mu} + n\bar{X} - \frac{n\mu}{2}.$$

Taking the derivative:

$$\frac{d\ell}{d\mu} = -\frac{n}{2\mu} + \frac{nm}{2\mu^2} - \frac{n}{2} = 0.$$

Multiplying by $2\mu^2/n$:

$$-\mu + m - \mu^2 = 0 \quad \Rightarrow \quad \mu^2 + \mu - m = 0.$$

By the quadratic formula: $\mu = \frac{-1\pm\sqrt{1+4m}}{2}$. Since $\mu > 0$ (as variance must be positive), we take the positive root:

$$\boxed{\hat{\mu} = \frac{-1+\sqrt{1+4m}}{2}.}$$

(b) Given that $\sigma = \mu$, show that the MLE of μ is $\hat{\mu} = \frac{-\bar{X}+\sqrt{\bar{X}^2+4m}}{2}$.

Proof. When $\sigma = \mu$, we have $\sigma^2 = \mu^2$. The log-likelihood is

$$\ell(\mu) = -\frac{n}{2}\log(2\pi) - n\log\mu - \frac{1}{2\mu^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Expanding:

$$\ell(\mu) = -\frac{n}{2}\log(2\pi) - n\log\mu - \frac{nm}{2\mu^2} + \frac{n\bar{X}}{\mu} - \frac{n}{2}.$$

Taking the derivative:

$$\frac{d\ell}{d\mu} = -\frac{n}{\mu} + \frac{nm}{\mu^3} - \frac{n\bar{X}}{\mu^2} = 0.$$

Multiplying by μ^3/n:

$$-\mu^2 + m - \bar{X}\mu = 0 \quad \Rightarrow \quad \mu^2 + \bar{X}\mu - m = 0.$$

By the quadratic formula: $\mu = \frac{-\bar{X}\pm\sqrt{\bar{X}^2+4m}}{2}$. Taking the positive root:

$$\boxed{\hat{\mu} = \frac{-\bar{X}+\sqrt{\bar{X}^2+4m}}{2}.}$$

Problem 3

Let $X_1, \dots, X_n$ be a random sample from a $N(\mu, 1)$ distribution.

(a) Show that $T = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - 1$ is an unbiased estimator of μ^2.

Proof. For $X_i \sim N(\mu, 1)$, we have $\mathrm{E}[X_i^2] = \mathrm{Var}(X_i) + (\mathrm{E}[X_i])^2 = 1 + \mu^2$. Therefore,

$$\mathrm{E}[T] = \mathrm{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i^2 - 1\right] = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}[X_i^2] - 1 = (1 + \mu^2) - 1 = \mu^2.$$

Hence T is unbiased for μ^2. □

(b) Explain why $2T$ is not an appropriate estimator of $2\mu^2$.

Solution. Although $\mathrm{E}[2T] = 2\mu^2$, the estimator $2T$ is not appropriate because T (and hence $2T$) can take negative values. Since $2\mu^2 \geq 0$ for all μ, an estimator that can be negative violates the natural constraint of the parameter space. For instance, if $\mu = 0$ and $n = 1$, then $T = X_1^2 - 1$ has a positive probability of being negative when $|X_1| < 1$.

(c) Give a simple proof that T is a biased estimator of μ.

Proof. From part (a), $\mathrm{E}[T] = \mu^2$. For T to be unbiased for μ, we would need $\mathrm{E}[T] = \mu$, i.e., $\mu^2 = \mu$ for all μ. This only holds when $\mu \in \{0, 1\}$. For any other value of μ, we have $\mathrm{E}[T] = \mu^2 \neq \mu$, so T is biased for μ. □

Problem 4

(a) Define what it means for T_n to be a consistent estimator of a parameter θ.

Definition. T_n is a consistent estimator of θ if $T_n \xrightarrow{p} \theta$ as $n \to \infty$, i.e., for any $\varepsilon > 0$,

$$P(|T_n - \theta| > \varepsilon) \to 0 \quad \text{as } n \to \infty.$$

(b) Given T_n is unbiased for θ, use Chebyshev's inequality to show that T_n is consistent for θ if $\mathrm{Var}(T_n) \to 0$ as $n \to \infty$.

Proof. By Chebyshev's inequality, for any $\varepsilon > 0$:

$$P(|T_n - \mathrm{E}[T_n]| > \varepsilon) \leq \frac{\mathrm{Var}(T_n)}{\varepsilon^2}.$$

Since T_n is unbiased, $\mathrm{E}[T_n] = \theta$. Thus:

$$P(|T_n - \theta| > \varepsilon) \leq \frac{\mathrm{Var}(T_n)}{\varepsilon^2} \to 0 \quad \text{as } n \to \infty,$$

provided $\mathrm{Var}(T_n) \to 0$. Therefore $T_n \xrightarrow{p} \theta$, i.e., T_n is consistent for θ. $\square$

Problem 5

Let $X_1, \dots, X_n$ be i.i.d. $N(\theta, 1)$ random variables.

(a) Obtain the CRLB for any unbiased estimator of θ.

Solution. The log-likelihood for one observation is

$$\log f(x;\theta) = -\frac{1}{2}\log(2\pi) - \frac{(x-\theta)^2}{2}.$$

The score function is $S_1(\theta) = \frac{\partial}{\partial\theta}\log f = x - \theta$.

The Fisher information for one observation is

$$I_1(\theta) = \mathrm{E}[(X-\theta)^2] = \mathrm{Var}(X) = 1.$$

For n i.i.d. observations: $I_n(\theta) = nI_1(\theta) = n$.

The CRLB for any unbiased estimator T of θ is:

$$\boxed{\mathrm{Var}(T) \geq \frac{1}{I_n(\theta)} = \frac{1}{n}.}$$

(b) By writing the score function in suitable form, identify the function $\tau(\theta)$ for which an MVBU estimator exists. Write down the latter estimator.

Solution. The total score function is

$$S(\theta) = \sum_{i=1}^{n}(X_i - \theta) = n\bar{X} - n\theta = n(\bar{X} - \theta).$$

This can be written as

$$S(\theta) = I_n(\theta) \cdot (\bar{X} - \theta),$$

which has the form $S(\theta) = I_n(\theta) \cdot (\hat{\tau} - \tau(\theta))$ with $\tau(\theta) = \theta$ and $\hat{\tau} = \bar{X}$.

By the Cram'er-Rao theorem, MVBU exists for $\tau(\theta) = \theta$, and the MVBU estimator is $\boxed{\bar{X}}$.

(c) Write down the UMVUE for θ.

Solution. Since $\bar{X}$ achieves the CRLB, it is the MVBU (minimum variance bound unbiased) estimator and hence is the UMVUE for θ:

$$\boxed{\text{UMVUE} = \bar{X}.}$$

(d) Obtain the CRLB for any unbiased estimator of 2θ.

Solution. Let $\tau(\theta) = 2\theta$. Then $\tau'(\theta) = 2$. By the Cram'er-Rao inequality:

$$\mathrm{Var}(T) \geq \frac{[\tau'(\theta)]^2}{I_n(\theta)} = \frac{4}{n}.$$

$$\boxed{\text{CRLB for } 2\theta = \frac{4}{n}.}$$

(e) By writing the score function in suitable form, identify the MVBU estimator of 2θ if it exists.

Solution. The score function is $S(\theta) = n(\bar{X} - \theta)$. For $\tau(\theta) = 2\theta$ with $\tau'(\theta) = 2$:

$$S(\theta) = n(\bar{X} - \theta) = \frac{n}{2}(2\bar{X} - 2\theta) = \frac{I_n(\theta)}{\tau'(\theta)}(\hat{\tau} - \tau(\theta)),$$

where $\hat{\tau} = 2\bar{X}$.

This matches the form required for MVBU existence. Therefore, the MVBU estimator of 2θ is $\boxed{2\bar{X}}$.

We can verify: $\mathrm{Var}(2\bar{X}) = 4\,\mathrm{Var}(\bar{X}) = 4/n$, which equals the CRLB.

Problem 6

Let $X_1, \dots, X_n$ be i.i.d. Poisson(λ) random variables. We wish to estimate $p = P(X_j = 0)$.

(a) Find an expression for p in terms of λ.

Solution. For $X \sim \text{Poisson}(\lambda)$:

$$\boxed{p = P(X=0) = \frac{e^{-\lambda}\lambda^0}{0!} = e^{-\lambda}.}$$

(b) Find the CRLB in terms of p for any unbiased estimator of p.

Solution. The Fisher information for one Poisson observation is $I_1(\lambda) = 1/\lambda$, so $I_n(\lambda) = n/\lambda$.
Since $p = e^{-\lambda}$, we have $\frac{dp}{d\lambda} = -e^{-\lambda} = -p$.
By the Cram'er-Rao inequality:

$$\text{Var}(T) \geq \frac{(dp/d\lambda)^2}{I_n(\lambda)} = \frac{p^2}{n/\lambda} = \frac{p^2\lambda}{n}.$$

Since $\lambda = -\log p$, the CRLB in terms of p is:

$$\boxed{\text{CRLB} = \frac{-p^2 \log p}{n}.}$$

(c) Let $Y_j = \begin{cases} 1, & \textbf{if } X_j = 0, \\ 0, & \textbf{otherwise.} \end{cases}$ Find $\mathrm{E}\, Y_j$ and use this result to write down an unbiased estimator T of p.

Solution. We have

$$\mathrm{E}[Y_j] = P(X_j = 0) = e^{-\lambda} = p.$$

Therefore, an unbiased estimator of p is

$$\boxed{T = \bar{Y} = \frac{1}{n}\sum_{j=1}^{n} Y_j.}$$

(d) Find $\text{Var}(T)$ (in terms of p).

Solution. Since $Y_j \sim \text{Bernoulli}(p)$, we have $\text{Var}(Y_j) = p(1-p)$. Therefore,

$$\boxed{\text{Var}(T) = \text{Var}(\bar{Y}) = \frac{\text{Var}(Y_1)}{n} = \frac{p(1-p)}{n}.}$$

(e) By graphing suitable functions or otherwise, determine if T is the MVBU estimator.

Solution. T is MVBU if and only if $\text{Var}(T) = \text{CRLB}$, i.e.,

$$\frac{p(1-p)}{n} = \frac{-p^2 \log p}{n}.$$

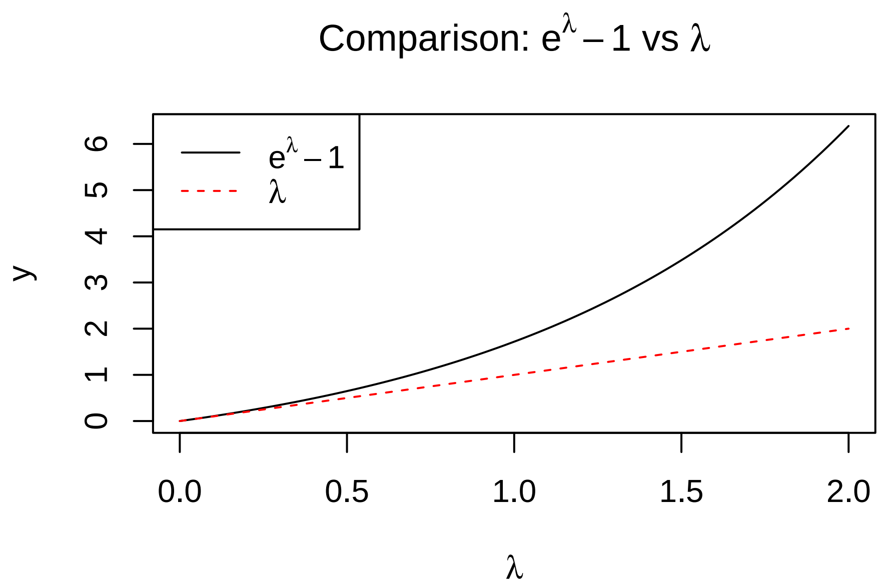
This simplifies to:

$$1 - p = -p \log p \quad \Leftrightarrow \quad \frac{1-p}{p} = -\log p \quad \Leftrightarrow \quad \frac{1}{p} - 1 = -\log p.$$

Let $\lambda = -\log p$ (so $p = e^{-\lambda}$). The condition becomes:

$$e^{\lambda} - 1 = \lambda.$$

```r
lambda_vals <- seq(0, 2, length.out = 100)
plot(lambda_vals, exp(lambda_vals) - 1, type = "l",
     ylab = "y", xlab = expression(lambda),
     main = expression(paste("Comparison: ", e^lambda - 1, " vs ", lambda)))
lines(lambda_vals, lambda_vals, lty = 2, col = "red")
legend("topleft", legend = c(expression(e^lambda - 1), expression(lambda)),
       lty = c(1, 2), col = c("black", "red"))
```



The curves intersect only at $\lambda = 0$ (i.e., $p = 1$), which is a degenerate case. For all $\lambda > 0$, we have $e^{\lambda} - 1 > \lambda$ by the strict convexity of e^{λ}.

This means $\text{Var}(T) > \text{CRLB}$ for all $p \in (0, 1)$. Therefore, T does **not** achieve the CRLB and is **not** the MVBU estimator.